

Multi-Layer LiDAR Sensor Fusion for SEPTA Light-Rail Operations

Passive hazard detection, route-region segmentation, and safety assurance

Proposed for Philadelphia trolley corridors - initial focus: Route 34 surface segment

2	6-9 mo	0	<200 ms	IV&V
POC VEHICLES	FIELD WINDOW	BRAKE AUTHORITY	P95 LATENCY TARGET	INDEPENDENT GATE

PREPARED BY

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TARGET AGENCY

Southeastern Pennsylvania Transportation Authority (SEPTA)

GEOGRAPHY

Philadelphia and Delaware County, Pennsylvania

DATE / STATUS

July 2026 - Pre-pilot proposal - Not agency-approved

Executive Summary

This proposal adapts fw/r's multi-sensor perception concept to SEPTA's trolley operating environment. SEPTA identifies eight trolley lines totaling 68 miles; its current vehicles date from the early 1980s, while 130 new low-floor trolleys are contracted for delivery beginning in 2027. [S1][S2] The recommended first step is a **two-vehicle, passive-only POC** on the Route 34 surface segment between 40th Street Portal and 61st Street/Baltimore. SEPTA's schedule and map identify that service as in-street trolley operation. [S3][S4]

The POC would log synchronized LiDAR, radar, camera, and vehicle-state data; detect vulnerable road users and track obstructions; and present advisory information to trained evaluators. It would have **no command path to propulsion or braking**. Deployment would proceed only through SEPTA's Safety Risk Management process, PennDOT State Safety Oversight review as applicable, OEM engineering approval, roadway-worker protections, and an independent verification gate. [S5][S6][S7]

2 LRVs INSTRUMENTED PILOT Proposed assumption; final vehicle selection by SEPTA.	6-9 mo FIELD WINDOW Includes design, installation, logging, and closeout.	Passive INITIAL MODE No automatic intervention or safety credit claimed.
\$1.30-2.05M ROM RANGE Non-binding Class 4-style planning range.	Gate-based DECISION MODEL Expand only after acceptance evidence and safety approval.	Route 34 STARTING CORRIDOR Proposed; subject to operational and community review.

Core proposition: create a SEPTA-owned evidence package for mixed-traffic perception - measured against pre-agreed targets - without modifying the certified braking or train-control functions. The output is a go / revise / stop decision, not a claim of production readiness.

Philadelphia / SEPTA Operational Context

Verified network and program facts

Component	Verified detail	Source
SEPTA Metro trolley network	Routes 10, 11, 13, 15, 34, 36, 101, and 102 are included in SEPTA Metro.	[S1]
Network scale	Eight trolley lines; 68 route miles; service links Philadelphia and Delaware County with Center City and University City.	[S2]
Current fleet context	SEPTA says current trolley vehicles have served since the early 1980s.	[S2]
Fleet replacement	130 low-floor trolleys under contract; option for 30 more; deliveries scheduled to begin in 2027 and complete by end of 2030.	[S2][S8]
Route 34	13th Street to 61st-Baltimore / Angora via Baltimore Avenue; official map distinguishes in-street track from tunnel service.	[S3][S4]
Oversight scope	PennDOT SSOA covers the T, G, and D trolley lines and their stations / maintenance facilities.	[S5]

Safety and infrastructure context

Verified 2025 oversight record	Implication for this POC
PennDOT conducted risk-based inspections of embedded track after T Line derailments and the annual T Line trolley blitz. [S5]	Sensor installation and test plans must be operationally subordinate to track, vehicle, and service controls.
FTA Special Directives 25-1 and 25-2 addressed SEPTA overhead-catenary inspection and maintenance; PennDOT describes work at 40th Street Portal. [S5]	The pilot cannot burden catenary corrective work or imply LiDAR mitigates catenary risk.
PennDOT continued safety certification oversight for Trolley Modernization and tracked SEPTA corrective actions. [S5]	The POC safety case should interface with existing Safety Risk Management and certification governance.

Fact / proposal boundary: The Route 34 focus, two-vehicle scope, sensor suite, schedule, cost, and all performance thresholds are proposed assumptions. No SEPTA incident baseline, stopping curve, vehicle interface, or route-specific risk ranking was available for this proposal; those are Phase 0 data requests.

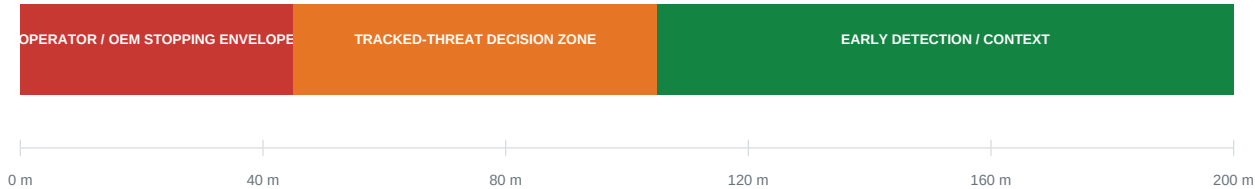
Operating Design Domain & Safety Envelope

The initial operating design domain (ODD) is intentionally narrow. It defines where evidence may be collected; it does not confer authority to control the vehicle.

ODD dimension	Proposed inclusion	Explicit exclusion / gate
Geography	Route 34 surface track, 40th Street Portal to 61st Street/Baltimore	Center City tunnel testing until separately approved
Vehicle mode	Revenue service logging or protected non-revenue test runs, as SEPTA directs	No change to operator rules, braking, propulsion, or signal compliance
Conditions	Day / night, dry / wet, seasonal glare, normal traffic	Severe weather or work zones unless specifically released
Objects	Pedestrians, cyclists, micromobility, road vehicles, track obstructions	Identity recognition, facial recognition, passenger surveillance
Outputs	Time-stamped detections, tracks, alerts to evaluator, health telemetry	Safety-critical command, automatic brake demand, public-facing alert

Decision distance is measured, not assumed

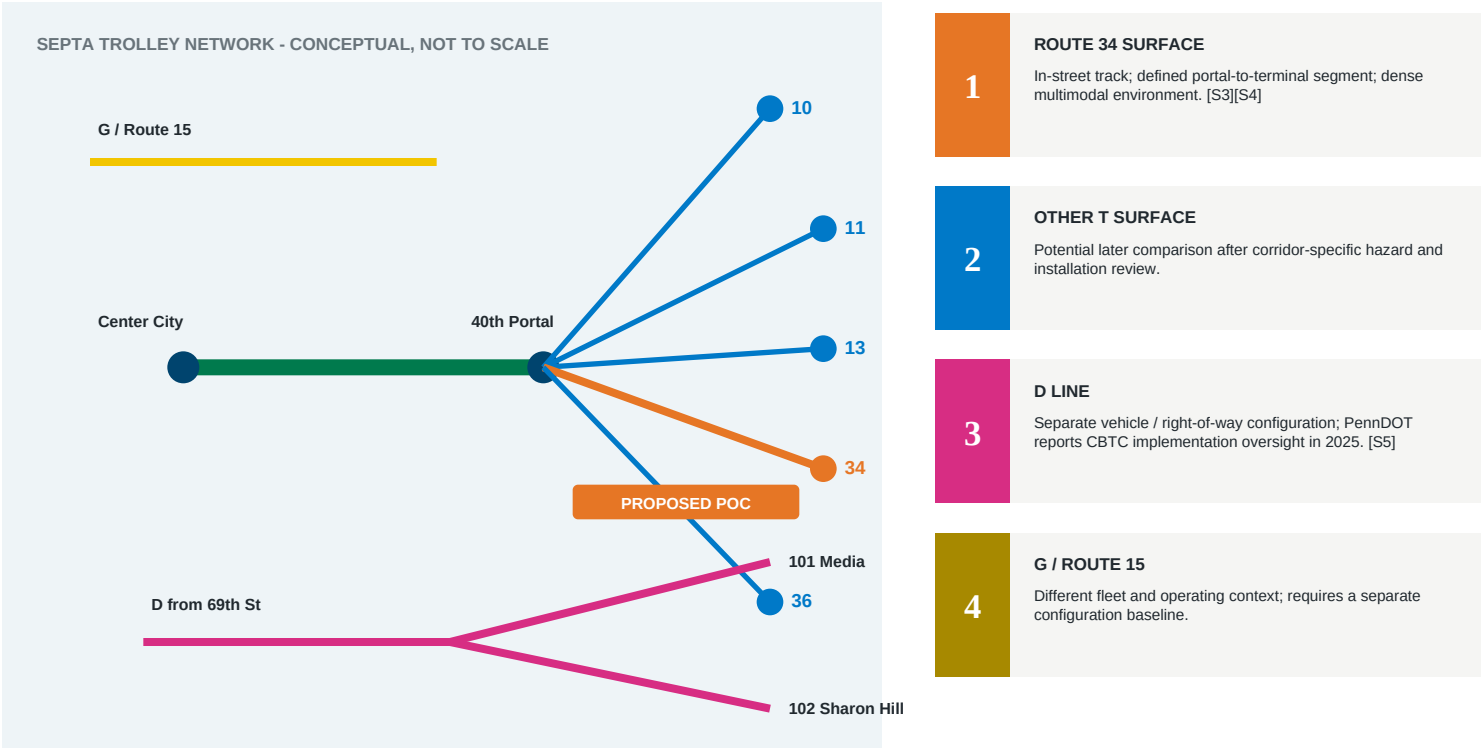
ILLUSTRATIVE - distances are placeholders until vehicle stopping tests / OEM curves are supplied



Gate A requires SEPTA-approved vehicle-specific stopping curves, maximum test speeds, grade, adhesion, load, operator reaction policy, sensor latency, and installation geometry. Until then, range graphics are conceptual and no stopping-margin claim is made.

Network Schematic & Deployment Priority Corridors

Route 34 is proposed as the first evidence corridor because the official SEPTA map shows a continuous in-street segment west of 40th Street Portal. The ranking below is a planning hypothesis, not a SEPTA risk determination. [S3][S4]



Phase 0 corridor data needed before final selection

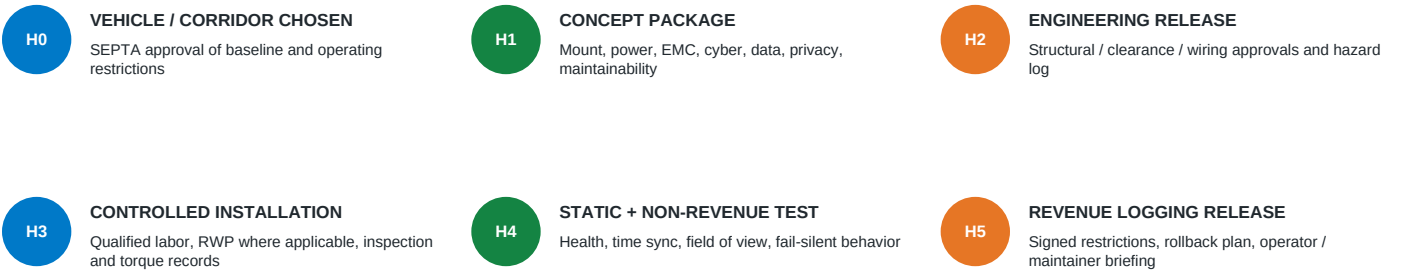
Evidence request	Owner / source	Why it matters
Five-year safety events, near misses, operator reports	SEPTA Safety / NTD source records	Establish corridor baseline without speculative incident counts
Vehicle assignments, duty cycles, stopping curves	SEPTA Operations / Vehicle Engineering / OEM	Define test speeds, mounting, power, and safe decision envelope
Track, OCS, work zones, clearance envelope	SEPTA Infrastructure / PennDOT SSOA coordination	Avoid conflicts and identify protected installation windows
Community and accessibility input	SEPTA Engagement / City / riders	Confirm privacy, curb access, station, and equity requirements

Corridor Selection & Installation Baseline

A production-quality POC begins with configuration control. Each proposed corridor is scored only after verified agency data are available; this page defines the selection rubric and physical installation hold points.

Criterion	Weight	Evidence required	Pass condition
Hazard relevance	25%	Safety events, near misses, operator reports	Documented recurring exposure addressable by perception
Operational representativeness	20%	Speeds, traffic mix, stops, grades, duty cycles	Covers intended future operating domain
Installation feasibility	20%	Clearance, structure, OCS, power, EMC, ingress	OEM / SEPTA concept approval; no clearance conflict
Data quality	15%	Sight distance, occlusion, GNSS, lighting, weather	Sufficient coverage for labeled ground truth
Service / community burden	10%	Access, curb use, detours, privacy review	No material service degradation; engagement plan
Validation logistics	10%	Safe access, test windows, independent observer	Repeatable test plan and auditable records

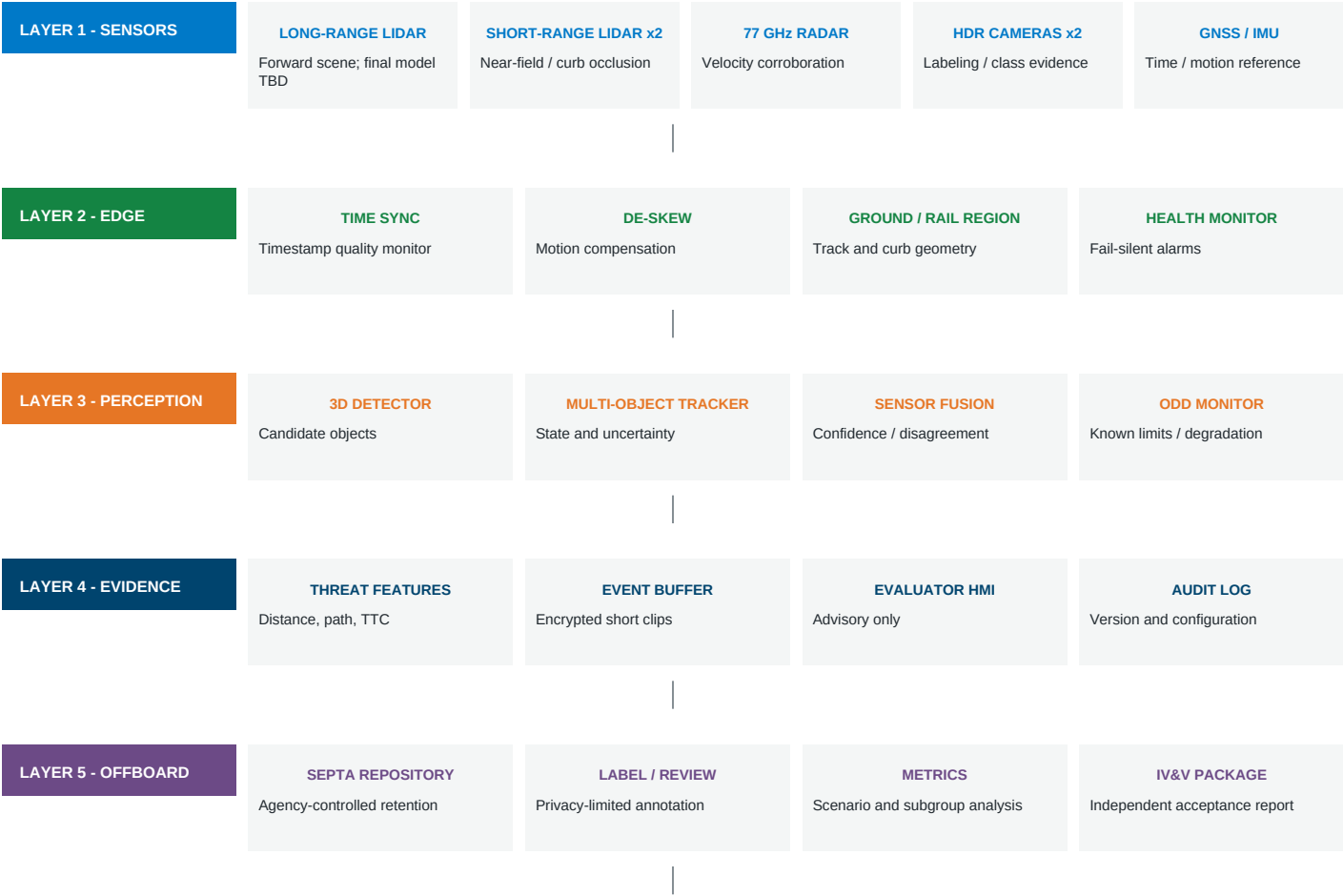
Vehicle installation hold points



No roof, cab, underframe, or apron location is assumed. Final sensor placement depends on the selected legacy vehicle or new trolley configuration, OEM restrictions, clearance envelope, pantograph / OCS interaction, sight lines, maintainability, and vandal resistance.

System Architecture

The proposed architecture is fail-silent and observational in Phase 1. A hardware data diode or equivalent engineered separation prevents the POC compute stack from commanding certified vehicle functions.



Phase 1 boundary: no brake command, no propulsion command, no train-control integration, no safety credit, no facial recognition, and no general-purpose video retention. Any future intervention concept is a new project requiring a separate hazard analysis, safety certification path, OEM design, cybersecurity review, and PennDOT SSOA coordination.

Detection Performance Analysis

The source document's performance figures are not carried forward as facts. SEPTA-specific performance must be measured on the selected vehicle and corridor. The graphics below are **proposed acceptance thresholds**, not achieved results.

PROPOSED PRIMARY ACCEPTANCE THRESHOLDS



PROPOSED LATENCY BUDGET - P95



Validation design

Evidence slice	Proposed minimum	Required reporting
Critical VRU encounters	250 independently adjudicated events	Recall with 95% confidence interval; day/night and wet/dry slices
Non-hazard background	100 vehicle-hours	False advisory rate per vehicle-hour and scenario
Controlled obstruction tests	100 runs across distance / occlusion matrix	Range-conditioned recall, localization error, time-to-detect
System health	Entire field window	Availability, dropped frames, time-sync error, thermal and storage limits
Human factors	At least 12 trained operators / evaluators if SEPTA approves	Usability, workload, trust calibration; no disciplinary use

Acceptance requires all safety-critical thresholds, no unresolved high-risk hazard, reproducibility on held-out runs, and independent review. A strong result authorizes only the next engineering phase - not automatic braking or revenue safety credit.

Implementation Roadmap

Four gated phases align the work with SEPTA engineering, Safety Risk Management, service windows, PennDOT SSOA coordination, and Trolley Modernization. Durations and costs are proposal assumptions.

PHASE 01	Discovery + Safety Basis Months 1-2 / \$180k-\$280k ROM
Confirm corridor, vehicles, governance, hazards, privacy, data rights, installation feasibility, and test method. ● Approve Phase 0 data request and configuration baseline ● Create hazard log, system boundary, cybersecurity / privacy plan ● Complete OEM / SEPTA concept reviews and PennDOT engagement plan	
PHASE 02	Two-Vehicle Passive Pilot Months 3-6 / \$620k-\$980k ROM
Install a fail-silent sensing stack on two selected vehicles. Conduct static, protected non-revenue, then approved revenue logging. ● Instrument, inspect, and commission two vehicles ● Collect representative day / night and dry / wet evidence ● No command path to brakes, propulsion, signals, or OCC safety functions	
PHASE 03	Independent Validation + Decision Months 7-9 / \$500k-\$790k ROM
Freeze the model, run held-out tests, assess human factors and maintainability, and issue a go / revise / stop package. ● Independent ground-truth review and reproducibility tests ● Close hazards or document residual risk owners ● Deliver SEPTA-controlled dataset, model card, configuration and final report	

Scale Path & Gated Schedule

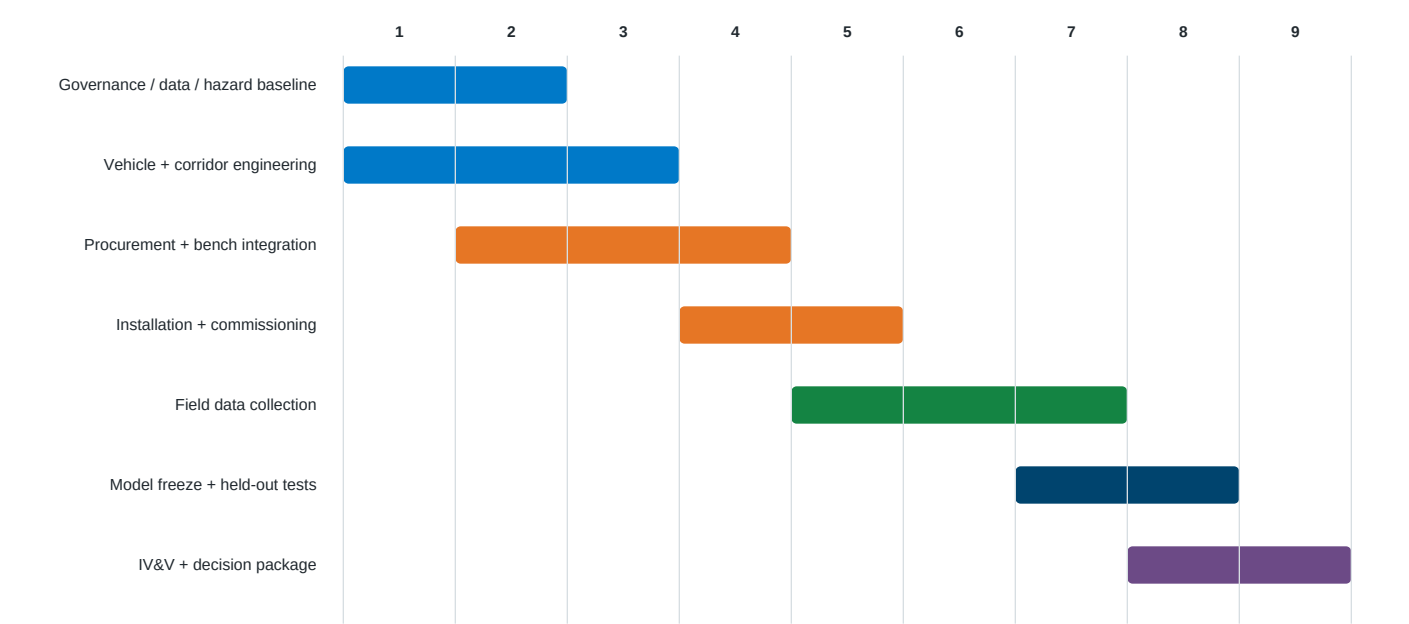
PHASE
04

Optional Pre-Production Engineering
Post-POC; separately authorized / Not included in POC ROM

If - and only if - Phase 3 passes, design a production architecture for selected legacy vehicles and/or the new low-floor fleet. [S2][S8]

- Coordinate with SEPTA Trolley Modernization and Alstom
- Develop safety-critical architecture only under a new certification plan
- Pilot fleet expansion and any intervention function require new approval

Nine-month illustrative Gantt



Schedule drivers: vehicle availability; OEM and structural review; SEPTA procurement; safe access / RWP; track and catenary work windows; privacy and labor consultation; representative weather; data labeling; and independent validation. No grant award, capital-program slot, or 2027 new-fleet access is assumed.

Stakeholder Map

PRIMARY - OWNER / OPERATOR SEPTA	Owns operational need, vehicle access, Agency Safety Plan / SMS processes, data, procurement, service rules, labor engagement, and decision authority. [S6]
PRIMARY - DEVELOPER fw/r Research	Proposes system architecture, integration engineering, model development, configuration records, and evidence package. No authority over SEPTA operations.
VEHICLE ENGINEERING SEPTA + vehicle OEM	Approve mounting, power, wiring, clearance, maintainability, warranty, electromagnetic compatibility, and future fleet interfaces.
NEW FLEET OEM Alstom	Supplier of 130 contracted low-floor Citadis vehicles, with 30 options; a future production interface is separate from the legacy-fleet POC. [S8]
STATE SAFETY OVERSIGHT PennDOT SSOA	Oversees SEPTA rail fixed-guideway safety, ASP approval, safety certification, inspections, investigations, and corrective actions under Part 674. [S5][S7]
FEDERAL FTA / US DOT	Part 673 PTASP / SMS, Part 674 state oversight, Part 671 roadway-worker protection, and applicable grant / procurement requirements. [S6][S7][S9]
LOCAL / ROADWAY City of Philadelphia + utilities	Coordinate street occupancy, curb / bike / pedestrian interfaces, traffic control, permits, and emergency access where applicable.
WORKFORCE Operators, maintainers, labor representatives	Co-design HMI and work practices; validate workload and trust; protect due process; prohibit covert performance monitoring.
INDEPENDENT ASSURANCE Qualified IV&V partner	Own test adjudication, statistical analysis, configuration audit, cybersecurity / privacy verification, and acceptance recommendation.

Financial Analysis & Cost-Benefit

All figures on this page are **non-binding POC planning assumptions** in 2026 dollars. They exclude vehicle acquisition, track / catenary renewal, production brake integration, and fleet rollout.

\$260k-\$380k

SENSORS + EDGE

Two vehicle kits, spares, bench system

\$300k-\$450k

ENGINEERING + INSTALL

Design, reviews, labor, commissioning

\$250k-\$400k

DATA + MODEL

Secure data ops, labeling, tuning

\$180k-\$300k

SAFETY + IV&V

Hazard process, test, independent review

\$310k-\$520k

PM + CONTINGENCY

Program, field support, 25% uncertainty

\$1.30M-\$2.05M

TOTAL POC ROM

Decision-quality evidence; no rollout

Cost-benefit method - no benefits claimed before baseline data

Input	Needed from SEPTA	Treatment
Addressable safety events / near misses	Route-level coded history and operator reports	Classify which scenarios perception could plausibly affect
Event consequence cost	Injury, property, legal, service disruption, investigation	Use SEPTA-approved avoided-cost method; sensitivity ranges
Reliability / service impacts	Delay minutes, road calls, maintenance burden	Count only measured deltas attributable to the system
Lifecycle cost	Hardware refresh, calibration, storage, cyber, training, spares	Discounted lifecycle model with fleet and ODD scenarios
Uncertainty	Detection efficacy, adoption, cost escalation, residual risk	Monte Carlo / scenario analysis; report downside case

Decision rule: Phase 3 compares the value of additional evidence and plausible lifecycle benefit with the cost and residual risk of a production program. The proposal does not reproduce the source document's NPV or payback figures because they are not verified for SEPTA.

Risk Register

Risk	Rating	Mitigation / owner
Pilot distracts from track / OCS corrective work or is misread as a substitute	HIGH	Sequence around SEPTA / PennDOT priorities; explicit non-substitution statement; SEPTA Safety owns release. [S5]
Vehicle mounting, clearance, EMC, power, or warranty conflict	HIGH	No assumed location; OEM + SEPTA engineering package; hold-point inspections; rollback configuration.
False reassurance from unverified performance claims	HIGH	Targets labeled as proposed; held-out tests; confidence intervals; independent report; no safety credit in Phase 1.
Privacy, surveillance perception, or inappropriate data use	HIGH	No facial recognition; minimize / encrypt / expire imagery; defined purpose; community and labor review; audit access.
Operator distraction or automation bias	MEDIUM-HIGH	Evaluator-only HMI first; human-factors protocol; alert suppression; operator retains full authority.
Cyber path reaches certified vehicle functions	HIGH	Engineered separation / data diode; no write-capable vehicle interface; penetration and architecture review.
Insufficient rare-event sample	MEDIUM	Scenario-based controlled tests plus field exposure; pre-set minimum sample; extend only by approved change.
Domain shift: glare, precipitation, occlusion, construction, seasonal change	MEDIUM	ODD monitor; stratified metrics; versioned retraining; out-of-domain fail-silent behavior.
RWP / installation safety exposure	HIGH	Comply with SEPTA rules and 49 CFR Part 671; qualified workers, job briefings, redundant protections where required. [S9]
Schedule / cost growth	MEDIUM	Stage-gated procurement, 25% ROM uncertainty, early vehicle survey, configuration freeze, explicit stop criteria.
New fleet interface changes before scale-up	MEDIUM	Treat Alstom fleet as a new configuration and certification basis; no transfer assumption. [S2][S8]

Success Metrics & Acceptance Criteria

Metric	Baseline	Proposed Phase 1 target	Measurement / decision
Critical-zone VRU recall	To be measured	At least 95%; lower 95% confidence bound reported	Independent event adjudication; day/night and wet/dry slices
Track obstruction recall	To be measured	At least 97% in controlled matrix	Distance / occlusion matrix; held-out tests
False advisories	To be measured	No more than 1.0 per vehicle-hour; severity-weighted review	100+ vehicle-hours; scenario taxonomy
End-to-end latency	Not available	P95 <= 200 ms; P99 <= 250 ms	Hardware timestamps; dropped-frame accounting
System availability	Not available	>= 98% during eligible logging hours	Health telemetry; planned downtime excluded by rule
Time synchronization	Not available	95% within 10 ms; no critical sample above 25 ms	Clock audit against recorded reference
Privacy / retention	Current policy review needed	100% compliance with approved purpose, access, redaction, retention	Access logs; sample audit; deletion certificate
Cyber separation	Not applicable	No write path to vehicle safety / control networks	Architecture inspection and penetration test
Operator / maintainer usability	To be measured	>= 80% rate workflow acceptable; no unresolved high-severity issue	Structured study; comments retained verbatim
Configuration reproducibility	Not applicable	100% of accepted runs traceable to software, model, calibration, and sensor serials	Independent configuration audit
Safety closure	Initial hazard log	No open unacceptable risk; residual risks assigned and accepted by authorized owner	SEPTA SRM records; PennDOT coordination as applicable

A threshold may not be waived after results are known. Any change requires documented rationale, impact analysis, approval, and re-test. Passing these gates supports a recommendation for further engineering only.

Conclusion & Recommended Next Steps

SEPTA's trolley network is a credible setting for a mixed-traffic perception POC: it comprises eight lines and 68 miles, includes in-street operation, and is entering a major vehicle and station modernization cycle. [S1][S2][S4] The most defensible near-term action is not automated braking. It is a tightly governed, passive-only pilot that produces corridor-specific evidence while respecting SEPTA's existing safety, track, catenary, vehicle, workforce, and regulatory priorities.

Route 34's surface segment is the proposed starting point, subject to SEPTA's own risk data and operational review. The POC uses two vehicles over six to nine months, a \$1.30M-\$2.05M ROM range, pre-registered acceptance gates, independent validation, and a stop option at every hold point. No current performance, avoided incidents, payback, stopping margin, or production readiness is claimed.

Recommended immediate next steps

STEP 01 30 DAYS	SEPTA engineering + safety working session Confirm business owner, safety owner, vehicle engineering lead, corridor hypothesis, and Phase 0 data access.
STEP 02 45 DAYS	PennDOT SSOA pre-application briefing Confirm expected SRM / safety-certification touchpoints and interface with ongoing trolley oversight. [S5]
STEP 03 60 DAYS	Vehicle and corridor survey Survey candidate vehicles and Route 34; define clearance, power, OCS, EMC, data, access, and rollback constraints.
STEP 04 75 DAYS	POC charter + test protocol Approve scope, privacy, labor engagement, acceptance statistics, independent reviewer, and stop / change rules.
STEP 05 90-120 DAYS	Competitive procurement / authorization Finalize statement of work, ROM basis, risk allocation, evidence ownership, and installation / test release.

Recommended decision: authorize a 60-day discovery and safety-basis phase only. Release hardware procurement and vehicle installation after the H1/H2 engineering gates are signed.

Sources, Verification & Assumptions

Sources are primary agency, regulator, and manufacturer materials. All links were checked July 11, 2026. Bracketed references [S#] map to this list. URLs are printed for human review.

S1 SEPTA, Get to Know SEPTA Metro Metro modes and trolley route list; terminology began February 2025. https://www.septa.org/metro/	S7 FTA, State Safety Oversight Program Certification 49 CFR Part 674 oversight framework. https://www.transit.dot.gov/regulations-and-guidance/safety/state-safety-oversight-program-certification-status
S2 SEPTA, New Trolley Fleet contract (updated Nov. 2025) Eight lines / 68 miles; 130 vehicles plus 30 options; schedule; current-fleet age; station program. https://www.septa.org/news/septa-board-awards-contract-for-purchase-of-new-trolley-fleet/	S8 Alstom, SEPTA Citadis contract announcement (June 1, 2023) Manufacturer confirmation of 130 low-floor electric vehicles and 30 options. https://www.alstom.com/press-releases-news/2023/6/alstom-selected-southeastern-pennsylvania-transportation-authority-deliver-130-low-floor-electric-citadis-light-rail-vehicles
S3 SEPTA, Route 34 schedule Route endpoints, Baltimore Avenue service, published service information. https://schedules.septa.org/current/34.pdf	S9 FTA, Rail Transit Roadway Worker Protection 49 CFR Part 671 applicability, program, training, and protection requirements. https://www.transit.dot.gov/regulations-and-programs/safety/rail-transit-roadway-worker-protection
S4 SEPTA, Subway-Surface Trolley map Tunnel, portal, and in-street track mapping. https://schedules.septa.org/maps/T4.pdf	S10 SEPTA, Trolley Modernization Program goals; accessible vehicles, stations, and operational / infrastructure improvements. https://www.septa.org/initiatives/trolley-modernization/
S5 PennDOT SSOA, 2025 Annual Safety and Security Report Oversight scope; 2025 inspections, directives, corrective actions, and trolley project oversight. https://www.pa.gov/content/dam/copapwp-pagov/en/pennDOT/documents/programs-and-doing-business/transit/documents/2025-ssoa-annual-report.pdf	S11 SEPTA, FY2027 Operating and Capital Budget overview Current capital context and trolley fleet-replacement investment. https://www.septa.org/about/budget/
S6 Federal Transit Administration, Public Transportation Agency Safety Plans 49 CFR Part 673 and Safety Management System requirements. https://www.transit.dot.gov/PTASP	S12 SEPTA, Reports portal Current route statistics, budgets, performance and program documents. https://www.septa.org/reports/

Verification boundary: Network, fleet, route, oversight, regulation, and modernization statements are cited facts. Corridor priority, vehicle count, duration, costs, architecture, ODD, performance thresholds, sample sizes, schedule, and next steps are proposal assumptions. No claim is made that SEPTA, PennDOT, FTA, Alstom, or the City has reviewed or endorsed this proposal.